VEENA CHIDANAND BADIGER

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PROFILE SUMMARY

Electronics Engineer with hands-on experience in building intelligent, real-time embedded and IoT systems, demonstrated through projects such as an AI-enabled Smart Electric Meter, a performance-optimized Dual- Axis Solar Tracker, and an autonomous Fire-Fighting Robot. Strong foundation in digital/embedded design, and low-power IoT architectures, complemented by practical skills in microcontrollers, sensors, control algorithms, and hardware–software integration to deliver efficient, data-driven engineering solutions.

RESEARCH INTERESTS

- **Low-Power SoC Architectures for IoT and Edge Devices:** Energy-efficient SoC design focusing on clock/power gating, low-leakage logic, and architecture-level optimization for next-generation smart IoT systems.
- **Low-Power Memory & Compute Architectures for AI Edge Processing:** Design of optimized SRAM/memory subsystems and lightweight compute units to support quantized neural networks, enabling fast and power-efficient AI inference at the edge.
- **FPGA-Based Edge AI Acceleration:** Development of real-time, low-power CNN and QNN hardware accelerators on FPGA platforms for tasks such as object detection and edge intelligence.

EDUCATION

Bachelors in Electronics and Instrumentation **Dec.'22 - Jun.'26**
JSS Academy of Technical Education, India (*National Assessment and Accreditation Council grade: A+)* **CGPA: 8.27**
Relevant Coursework: VLSI Design, Digital Signal Processing, Digital Design and HDL, Embedded Systems, Basic Electronics, IOT, Computer Organization and Architecture, Operating Systems, Computer Communication Networks, Machine Learning, Control System, Analog Electronic Circuits.

PROJECTS

Smart Electric Meter System **Aug. – Dec. '25**

- Developed an AI-enabled smart energy meter using ESP32, IoT sensors (current/voltage), MQTT/serial communication, Python, and a web dashboard, enabling real-time power monitoring, remote access (Telegram Bot API), and reliable data logging.
- Implemented automated energy-management logic (presence-based lighting and temperature-based fan control), along with future monthly consumption prediction and real-time device-level load classification, delivering an efficient, low-power IoT energy-management system.

Sun Tracking Solar Panel **Feb. – Jun. '24**

- Designed a dual-axis solar tracker using Arduino UNO, LDR sensors, servo motors, and real-time monitoring to dynamically adjust the panel toward maximum light intensity.
- Implemented a closed-loop tracking algorithm (Arduino C/C++), validating performance through real- time solar intensity and power measurements, achieving ~25% higher energy capture than fixed panels.

Fire Fighting Robot using Arduino UNO **Apr. – Jul. '23**

- Developed a real-time fire-detection and autonomous extinguishing robot using Arduino UNO, IR flame sensors, L298N motor driver, DC motors, servo mechanism, and a water-pump module, enabling accurate flame localization and automated suppression.
- Demonstrated reliable operation across varying flame intensities and distances, validating a robust fire- response prototype through consistent real-world performance tests.

PUBLICATIONS

Veena Chidanand Badiger, Shreya K R, Satish, Chirag H K S, B R Renukumar (2026). SMART ELECTRIC METER SYSTEM. International Journal of Innovative Research In Technology (IJIRT),12(8), 2184-2188.
Jan.'2026

This paper presents the design and implementation of a Smart Electric Meter System aimed at improving energy monitoring and management. The proposed system enables real-time measurement of electricity consumption, automated data transmission, and reduced manual intervention. By integrating embedded systems and communication technologies, the solution enhances billing accuracy, minimizes power theft, and supports efficient energy usage. The system demonstrates a cost-effective and scalable approach suitable for modern smart grid applications.
https://ijirt.org/article?manuscript=190146

INTERNSHIPS/TRAINING

PCB Design Intern **Jan.'26-May.'26**
SkillDzire Technologies Pvt. Ltd.
- • PCB schematic capture and layout using KiCad for mixed-signal circuits.
- • Designed PCBs for servo drivers, LED circuits, sensor interfaces, and RF applications.
- • Focused on signal integrity, power routing, and manufacturability.

Intern – FPGA & Embedded Systems
Bharat Electronics Limited (BEL), PDIC, Bengaluru **Feb.'26 – May.'26**
- Developed and tested an FPGA-based temperature-analysis system using TMP126 and FPGA processing modules.
- Worked on hardware–software interfacing, automation scripts, debugging, simulation and system validation.

VLSI Physical Design Trainee
Maven Silicon **July.'26 – Present**
- Hands-on training in ASIC/VLSI design flow, Verilog HDL, CMOS, synthesis, STA, Linux, and Physical Design fundamentals.
- Practicing RTL design and synthesis using Design Compiler and Tcl.

Engineering Intern – System Validation
Marelli **Sep.'26 – Present**
- Supporting testing and validation activities in an automotive electronics environment.
- Gaining hands-on experience in debugging, documentation, and structured testing workflows.

TECHNICAL SKILLS

- **Programming Languages**: C, Python, Verilog
- **Simulation & Design Tools**: Cadence Virtuoso, Quartus Prime, ModelSim, Xilinx ISE, Tanner S-Edit
- **Operating Systems:** Red Hat Linux, Windows
- **Other Tools:** Arduino IDE, LabVIEW, MATLAB, Keil µVision, Microsoft office, CANoe

CERTIFICATIONS

- ASIC Design & Verification using Verilog HDL + Project Demo – Udemy **Oct.'25**
  – RTL design, simulation, synthesis, and end-to-end ASIC design workflow.

- VLSI for Beginners – National Institute of Electronics & Information Technology (NIELIT), Calicut **May.'25**
  – Fundamentals of VLSI design, digital logic, and IC design basics.

- VLSI Design Flow: RTL to GDS – NPTEL **Jul-Oct.'2025**
  – Detailed understanding of the complete VLSI design flow including RTL coding, synthesis, floorplanning, placement, routing, and GDSII generation.

- Design Verification using System Verilog (SV) + Project Demo – Udemy **Oct.'26**
  – Fundamentals of SystemVerilog-based design verification, including testbench development, simulation, and verification methodologies.

- ASIC Synthesis-STA-Physical Design(PD):Cadence+Synopsys flow– Udemy **Oct.'26**
  – Hands-on understanding of ASIC synthesis, static timing analysis, and physical design using Cadence and Synopsys EDA tools.